

The Mass Flux (Robin) Boundary Condition in Finite Difference Solutions of the Advection–Dispersion Equation: Cell-Centred and Edge-Centred Formulations

1 Introduction

The most physically meaningful boundary condition (BC) for solute transport at inflow boundaries is the prescribed mass flux condition. It simply reflects that the fluid flux entering the system carries a known concentration c_{inf} , so that the total (advective plus dispersive) solute mass flux across the boundary equals $m_{\text{inf}} = q_{\text{inf}} c_{\text{inf}}$. In general, it is expressed as

$$(qc - D \nabla c) \cdot \mathbf{n} = -m_{\text{inf}}, \quad (1)$$

where q [L T^{-1}] is the fluid flux, D [$\text{L}^2 \text{T}^{-1}$] the hydrodynamic dispersion coefficient, c the transported scalar and $\mathbf{n}$ the outward unit normal; the sign follows the bankers criterion (inputs positive) customary in Earth Sciences ([Petchamé-Guerrero and Carrera, 2024](#)). In a one-dimensional domain $[0, L]$ with flow from left to right, the condition at the inflow boundary $x = 0$ reads

$$\left(qc - D \frac{\partial c}{\partial x} \right) \Big|_{x=0} = q_{\text{inf}} c_{\text{inf}}. \quad (2)$$

Since this condition prescribes a linear relationship between c and its gradient, it is a Robin (also called Cauchy) BC. At outflow boundaries ($x = L$) the mass flux may likewise be prescribed as $m_{\text{out}} = q_{\text{out}} c_{\text{out}}$,

$$\left(qc - D \frac{\partial c}{\partial x} \right) \Big|_{x=L} = q_{\text{out}} c_{\text{out}}. \quad (3)$$

In the most common case, it is assumed that the concentration leaving the domain is the resident concentration of the outflow cell, $c_{\text{out}} = c_n$, so that (3) reduces to the homogeneous Neumann condition $D \partial c / \partial x = 0$. In summary, mass flux BCs are of Robin type at inflow boundaries and (usually) of Neumann type at outflow boundaries.

We describe below the implementation of this BC for two finite difference (FD) spatial arrangements: (i) *cell-centred* FD, where unknowns are located at cell centres and the boundary coincides with a cell face, and (ii) *edge-centred* (node-centred) FD, where unknowns are located at nodes and the first and last nodes lie exactly on the boundaries. In both cases the BC is imposed by means of a dummy (ghost) cell or node, whose concentration is eliminated by enforcing the prescribed mass flux; for the improved scheme of [Petchamé-Guerrero and Carrera \(2024\)](#), the boundary flux can instead be substituted directly, without any dummy cell.

2 Governing Equations and Discretisation Framework

We consider the one-dimensional advection–dispersion equation (ADE) with constant porosity and density. Its advective form reads

$$\phi \frac{\partial c}{\partial t} = -q \frac{\partial c}{\partial x} + \frac{\partial}{\partial x} \left(D \frac{\partial c}{\partial x} \right) + r(c^* - c), \quad (4)$$

where $\phi [-]$ is the porosity, $r [\text{T}^{-1}]$ represents fluid sinks/sources and c^* is the concentration associated with them. Spatial discretisation on a uniform mesh of size Δx with n unknowns $c_1, \dots, c_n$ leads to the semi-discrete linear system

$$\phi \frac{dc}{dt} = \mathbf{T}\mathbf{c} + \mathbf{g}, \quad (5)$$

where $\mathbf{T}$ is the (tridiagonal) transport matrix and $\mathbf{g}$ collects boundary and sink/source contributions. Interior rows of $\mathbf{T}$ follow from the chosen scheme; boundary rows are obtained by eliminating the dummy concentrations with the discretised BC, as detailed below. Time is discretised in uniform steps Δt , so that superscript k denotes hereafter the time level $t^k = k \Delta t$. Any time-stepping scheme (θ -weighting, explicit Euler, Crank–Nicolson, etc.) with uniform time step can then be applied to (5).

Two spatial schemes are considered for the advective term (Petchamé-Guerrero and Carrera, 2024): the traditional centred FD scheme (TFDS), which approximates $q \partial c / \partial x$ at cell centres,

$$\left(q \frac{\partial c}{\partial x} \right) \Big|_i \approx q_i \frac{c_{i+1} - c_{i-1}}{2\Delta x}, \quad (6)$$

and the improved (mass-conserving) FD scheme (IFDS), which averages the advective term at the adjacent cell interfaces,

$$\left(q \frac{\partial c}{\partial x} \right) \Big|_i \approx \frac{1}{2} \left(q_{i+1/2} \frac{c_{i+1} - c_i}{\Delta x} + q_{i-1/2} \frac{c_i - c_{i-1}}{\Delta x} \right). \quad (7)$$

3 Cell-Centred Finite Differences

3.1 Grid arrangement and dummy cells

In the cell-centred arrangement, the domain is divided into n cells of uniform size Δx with unknowns at the cell centres, $x_i = (i - \frac{1}{2})\Delta x$, $i = 1, \dots, n$. The boundaries $x = 0$ and $x = L = n\Delta x$ coincide with the faces of the first and last cells. The BCs are imposed through dummy cells $i = -1$, centred at $x = -\Delta x/2$, and $i = n + 1$, centred at $L + \Delta x/2$ (Petchamé-Guerrero and Carrera, 2024). Note that no unknown is located on the boundary itself, so boundary values must be interpolated: the concentration at $x = 0$ is approximated as the arithmetic average $(c_{-1} + c_1)/2$, and the boundary gradient by the centred difference $(c_1 - c_{-1})/\Delta x$.

3.2 Inflow boundary

Discretising the mass flux condition (2) across the inflow face with these approximations yields, at time level k ,

$$m_0^k = q_{\text{inf}} c_{\text{inf}}^k = \left(qc - D \frac{\partial c}{\partial x} \right) \Big|_{\text{inf}} \approx q_{\text{inf}} \frac{c_{-1}^k + c_1^k}{2} + D_{\text{inf}} \frac{c_{-1}^k - c_1^k}{\Delta x}, \quad (8)$$

where D_{inf} is the dispersion coefficient at the inflow boundary. Solving for the dummy concentration gives

$$c_{-1}^k = \frac{q_{\text{inf}} \Delta x (2c_{\text{inf}}^k - c_1^k) + 2D_{\text{inf}} c_1^k}{q_{\text{inf}} \Delta x + 2D_{\text{inf}}} = c_1^k \frac{2D_{\text{inf}} - q_{\text{inf}} \Delta x}{q_{\text{inf}} \Delta x + 2D_{\text{inf}}} + c_{\text{inf}}^k \frac{2q_{\text{inf}} \Delta x}{q_{\text{inf}} \Delta x + 2D_{\text{inf}}}. \quad (9)$$

Equation (9) expresses the dummy concentration as a weighted combination of c_1 and c_{inf} , with weights controlled by the grid Péclet number $q_{\text{inf}} \Delta x / D_{\text{inf}}$. Its two limits are physically meaningful: when $q_{\text{inf}} = 0$ it reduces to $c_{-1} = c_1$ (zero dispersive flux across the boundary), whereas in the purely advective limit $D_{\text{inf}} \rightarrow 0$ it gives $c_{-1} = 2c_{\text{inf}} - c_1$, which is precisely the dummy-cell

reflection that prescribes the boundary concentration $(c_{-1} + c_1)/2 = c_{\text{inf}}$ (Dirichlet limit). Interestingly, this expression is rarely found in the literature, which is paradoxical given that the mass flux BC is the most physically meaningful condition at inflow boundaries (Petchamé-Guerrero and Carrera, 2024).

Traditional scheme (TFDS). Writing the traditional centred scheme (6) at the first cell, with the dispersive flux discretised using D_{inf} at the boundary face and $D_{3/2}$ at the interior face (the interface between cells 1 and 2),

$$\phi_1 \frac{dc_1}{dt} = -q_1 \frac{c_2 - c_{-1}}{2\Delta x} + D_{3/2} \frac{c_2 - c_1}{\Delta x^2} - D_{\text{inf}} \frac{c_1 - c_{-1}}{\Delta x^2} + r_1 (c_1^* - c_1), \quad (10)$$

and eliminating c_{-1} with (9) yields the first row of $\mathbf{T}$ and $\mathbf{g}$ in (5):

$$\begin{cases} T_{1,1} = \frac{2D_{\text{inf}} - q_{\text{inf}}\Delta x}{q_{\text{inf}}\Delta x + 2D_{\text{inf}}} \left(\frac{q_1\Delta x + 2D_{\text{inf}}}{2\Delta x^2} \right) - \frac{D_{3/2} + D_{\text{inf}}}{\Delta x^2} - r_1, \\ T_{1,2} = \frac{2D_{3/2} - q_1\Delta x}{2\Delta x^2}, \\ g_1 = \frac{q_{\text{inf}}c_{\text{inf}}}{q_{\text{inf}}\Delta x + 2D_{\text{inf}}} \left(\frac{q_1\Delta x + 2D_{\text{inf}}}{\Delta x} \right) + r_1c_1^*, \end{cases} \quad (11)$$

where q_1 is the fluid flux at the centre of the first cell and r_1, c_1^* the sink/source terms of that cell. Using D_{inf} in the boundary-face flux ensures a correct boundary mass balance: when the fluid flux is uniform near the boundary ($q_1 = q_{\text{inf}}$), the boundary contribution to g_1 reduces to $q_{\text{inf}}c_{\text{inf}}/\Delta x$, i.e., exactly the prescribed mass influx divided by the cell width.

Improved scheme (IFDS). Since the improved scheme evaluates advection at cell interfaces, the advective boundary flux $q_{\text{inf}}c_{\text{inf}}$ can be substituted *directly* into the interface flux balance of the first cell, without introducing any dummy cell (Petchamé-Guerrero and Carrera, 2024, Eq. (8)):

$$\phi_1 \frac{dc_1}{dt} = q_{\text{inf}} \frac{c_{\text{inf}} - c_1}{\Delta x} - q_{3/2} \frac{c_2 - c_1}{2\Delta x} + D_{3/2} \frac{c_2 - c_1}{\Delta x^2} + r_1 (c_1^* - c_1), \quad (12)$$

where $q_{3/2}$ and $D_{3/2}$ are evaluated at the interface between cells 1 and 2. The corresponding matrix row is

$$\begin{cases} T_{1,1} = -\frac{q_{\text{inf}}}{\Delta x} + \frac{q_{3/2}}{2\Delta x} - \frac{D_{3/2}}{\Delta x^2} - r_1, \\ T_{1,2} = \frac{D_{3/2}}{\Delta x^2} - \frac{q_{3/2}}{2\Delta x}, \\ g_1 = \frac{q_{\text{inf}}c_{\text{inf}}}{\Delta x} + r_1c_1^*. \end{cases} \quad (13)$$

Note that the boundary contribution to g_1 is simply the prescribed mass influx divided by the cell width: this guarantees exact mass conservation at the boundary by construction.

3.3 Outflow boundary

At the outflow face $x = L$, prescribing the mass flux $m_{\text{out}}^k = q_{\text{out}}c_{\text{out}}^k$ and discretising with the dummy cell $n + 1$,

$$m_{\text{out}}^k = q_{\text{out}}c_{\text{out}}^k = \left(qc - D \frac{\partial c}{\partial x} \right) \Big|_{\text{out}} \approx q_{\text{out}} \frac{c_{n+1}^k + c_n^k}{2} - D_{\text{out}} \frac{c_{n+1}^k - c_n^k}{\Delta x}, \quad (14)$$

where D_{out} is the dispersion coefficient at the outflow boundary. Solving for the dummy concentration gives

$$c_{n+1}^k = c_n^k \frac{2D_{\text{out}} + q_{\text{out}}\Delta x}{2D_{\text{out}} - q_{\text{out}}\Delta x} - c_{\text{out}}^k \frac{2q_{\text{out}}\Delta x}{2D_{\text{out}} - q_{\text{out}}\Delta x}. \quad (15)$$

Writing the traditional scheme at the last cell, with D_{out} at the boundary face and $D_{n-1/2}$ at the interior face (the interface between cells $n-1$ and n),

$$\phi_n \frac{dc_n}{dt} = -q_n \frac{c_{n+1} - c_{n-1}}{2\Delta x} + D_{\text{out}} \frac{c_{n+1} - c_n}{\Delta x^2} - D_{n-1/2} \frac{c_n - c_{n-1}}{\Delta x^2} + r_n (c_n^* - c_n), \quad (16)$$

and eliminating c_{n+1} with (15) yields the last row of the system:

$$\begin{cases} T_{n,n} = \frac{2D_{\text{out}} + q_{\text{out}}\Delta x}{2D_{\text{out}} - q_{\text{out}}\Delta x} \left(\frac{2D_{\text{out}} - q_n\Delta x}{2\Delta x^2} \right) - \frac{D_{n-1/2} + D_{\text{out}}}{\Delta x^2} - r_n, \\ T_{n,n-1} = \frac{q_n}{2\Delta x} + \frac{D_{n-1/2}}{\Delta x^2}, \\ g_n = -\frac{q_{\text{out}}c_{\text{out}}}{2D_{\text{out}} - q_{\text{out}}\Delta x} \left(\frac{2D_{\text{out}} - q_n\Delta x}{\Delta x} \right) + r_n c_n^*. \end{cases} \quad (17)$$

As at the inflow boundary, the mass balance is correct: when the fluid flux is uniform near the boundary ($q_n = q_{\text{out}}$), the boundary contribution to g_n reduces to $-q_{\text{out}}c_{\text{out}}/\Delta x$, i.e., the prescribed mass outflux (negative, following the bankers criterion) divided by the cell width.

In the usual case where the outflowing water carries the resident concentration ($c_{\text{out}} = c_n$, i.e., $D_{\text{out}} \partial c / \partial x = 0$), the implementation is much simpler: the dummy cell is just set to

$$c_{n+1}^k = c_n^k. \quad (18)$$

4 Edge-Centred Finite Differences

4.1 Grid arrangement and dummy nodes

In the edge-centred (node-centred) arrangement, the unknowns are located at nodes $x_i = (i-1)\Delta x$, $i = 1, \dots, n$, so that the first and last nodes lie *exactly on* the boundaries ($x_1 = 0$, $x_n = L$). This has an important consequence for the mass flux BC: the boundary concentration is a nodal unknown (c_1 at inflow), so the advective part of the boundary flux, $q_{\text{inf}}c_1$, is represented exactly and no interpolation is needed. Only the boundary gradient requires a dummy node $i = -1$ located at $x = -\Delta x$ (respectively $i = n+1$ at $x = L + \Delta x$), used to build the second-order centred difference $\partial c / \partial x|_{x=0} \approx (c_2 - c_{-1}) / (2\Delta x)$.

4.2 Inflow boundary

Discretising the mass flux condition (2) at node 1,

$$m_0^k = q_{\text{inf}}c_{\text{inf}}^k = \left(qc - D \frac{\partial c}{\partial x} \right) \Big|_{\text{inf}} \approx q_{\text{inf}}c_1^k + D_{\text{inf}} \frac{c_{-1}^k - c_2^k}{2\Delta x}. \quad (19)$$

Solving for the dummy concentration,

$$c_{-1}^k = c_{\text{inf}}^k \frac{2q_{\text{inf}}\Delta x}{D_{\text{inf}}} - c_1^k \frac{2q_{\text{inf}}\Delta x}{D_{\text{inf}}} + c_2^k = c_2^k + \frac{2q_{\text{inf}}\Delta x}{D_{\text{inf}}} (c_{\text{inf}}^k - c_1^k). \quad (20)$$

Compared with the cell-centred expression (9), the structure is quite different: the dummy concentration is anchored to c_2 (symmetry of the centred gradient) plus a correction proportional to the concentration deficit ($c_{\text{inf}} - c_1$) scaled by twice the grid Péclet number $q_{\text{inf}}\Delta x / D_{\text{inf}}$. Note also that (20) requires $D_{\text{inf}} > 0$: in the purely advective limit the edge-centred Robin condition degenerates into the Dirichlet condition $c_1 = c_{\text{inf}}$, which must then be imposed directly.

Matrix row. Writing the centred FD scheme at node 1,

$$\phi_1 \frac{dc_1}{dt} = -q_1 \frac{c_2 - c_{-1}}{2\Delta x} + D_1 \frac{c_2 - 2c_1 + c_{-1}}{\Delta x^2} + r_1 (c_1^* - c_1), \quad (21)$$

and eliminating c_{-1} with (20) (noting that, since node 1 lies exactly on the boundary, $D_{\text{inf}} = D_1$, the dispersion coefficient at that node) yields

$$\begin{cases} T_{1,1} = -\frac{2q_{\text{inf}}}{D_1} \left(\frac{q_1 \Delta x + 2D_1}{2\Delta x} \right) - \frac{2D_1}{\Delta x^2} - r_1, \\ T_{1,2} = \frac{2D_1}{\Delta x^2}, \\ g_1 = \frac{2q_{\text{inf}}}{D_1} \left(\frac{q_1 \Delta x + 2D_1}{2\Delta x} \right) c_{\text{inf}} + r_1 c_1^*. \end{cases} \quad (22)$$

Observe the elegant structure of (22): the boundary contributions to $T_{1,1}$ and g_1 are identical in magnitude and opposite in sign, so the boundary term enters the equation as $\frac{2q_{\text{inf}}}{D_1} \left(\frac{q_1 \Delta x + 2D_1}{2\Delta x} \right) (c_{\text{inf}} - c_1)$, i.e., a relaxation of c_1 towards c_{inf} whose rate grows with the inflow flux. Mass enters the domain only insofar as the boundary node has not yet reached the inflow concentration.

4.3 Outflow boundary

The same procedure applies at the outflow node $x_n = L$, using the dummy node $n+1$. Since node n lies exactly on the boundary, $D_{\text{out}} = D_n$, the dispersion coefficient at that node. Discretising (3),

$$m_{\text{out}}^k = q_{\text{out}} c_{\text{out}}^k \approx q_{\text{out}} c_n^k - D_n \frac{c_{n+1}^k - c_{n-1}^k}{2\Delta x} \implies c_{n+1}^k = c_{n-1}^k + \frac{2q_{\text{out}} \Delta x}{D_n} (c_n^k - c_{\text{out}}^k), \quad (23)$$

and eliminating c_{n+1} from the centred scheme at node n yields

$$\begin{cases} T_{n,n} = -\frac{2q_{\text{out}}}{D_n} \left(\frac{q_n \Delta x - 2D_n}{2\Delta x} \right) - \frac{2D_n}{\Delta x^2} - r_n, \\ T_{n,n-1} = \frac{2D_n}{\Delta x^2}, \\ g_n = \frac{2q_{\text{out}}}{D_n} \left(\frac{q_n \Delta x - 2D_n}{2\Delta x} \right) c_{\text{out}} + r_n c_n^*. \end{cases} \quad (24)$$

For the usual free-outflow condition ($c_{\text{out}} = c_n$, zero dispersive flux), (23) reduces to the simple reflection

$$c_{n+1}^k = c_{n-1}^k. \quad (25)$$

References

Petchamé-Guerrero, J., Carrera, J. (2024). An Improved Scheme for the Finite Difference Approximation of the Advective Term in the Heat or Solute Transport Equations. *Transport in Porous Media*, 151, 2795–2817. <https://doi.org/10.1007/s11242-024-02133-5>